

IMPLEMENTATION OF LOGIC FUNCTIONS

COMBINATIONAL LOGIC CIRCUITS

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TOPIC OUTLINE

Implementation of XOR/XNOR Gate

Implementation of Logic Functions

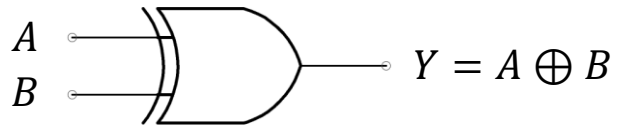


IMPLEMENTATION OF XOR/XNOR GATE



EXCLUSIVE-OR GATE

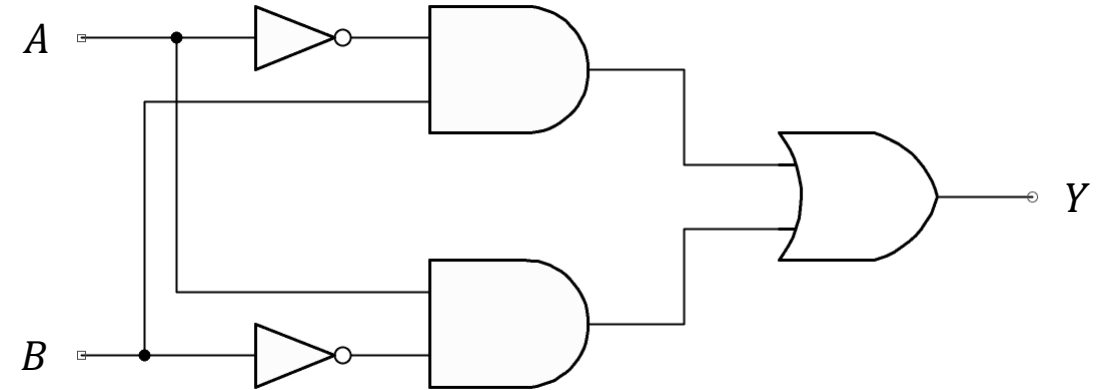
Logic Symbol



Truth Table

A	B	Y	Minterm
0	0	0	
0	1	1	
1	0	1	
1	1	0	

Equivalent Logic Circuit

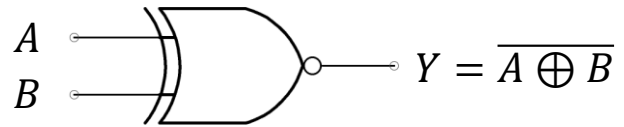


$$Y = \bar{A}B + A\bar{B}$$



EXCLUSIVE-NOR GATE

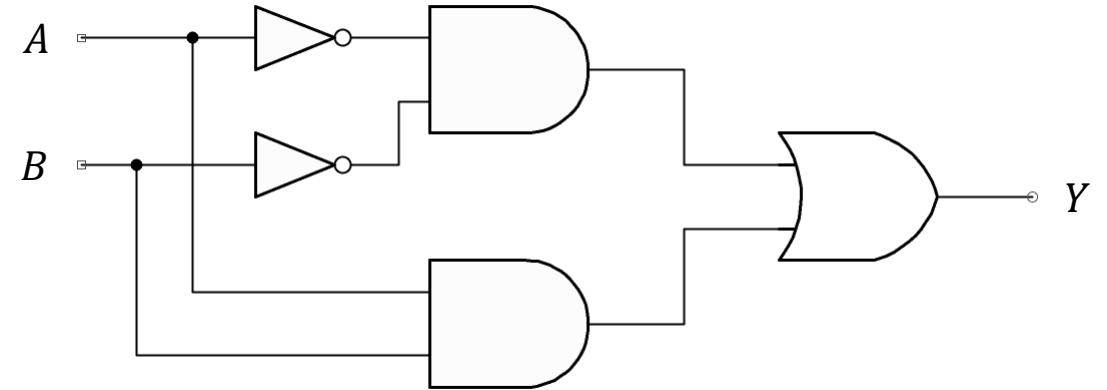
Logic Symbol



Truth Table

A	B	Y	Minterm
0	0	1	
0	1	0	
1	0	0	
1	1	1	

Equivalent Logic Circuit



IMPLEMENTATION OF LOGIC FUNCTIONS



EXERCISE

Implement the logic function describe by the truth table.

Solution

A	B	C	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



EXERCISE

A section of a bubble gumball factory uses a conveyor system equipped with three sensors— s_1 , s_2 , and s_3 to inspect each gumball.

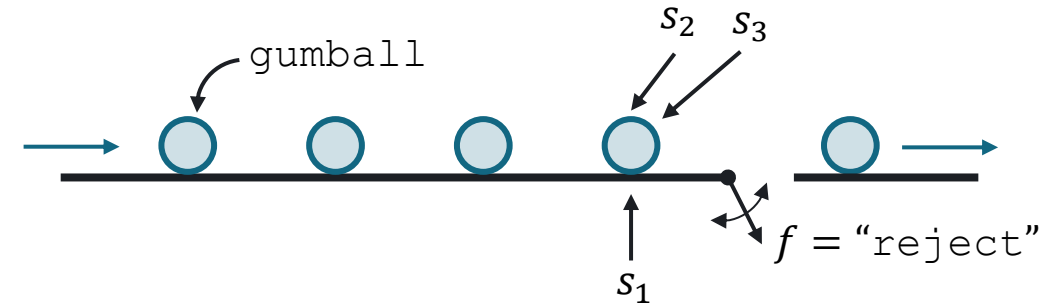
- $s_1 = 1$ if the gumball is too light.
- $s_2 = 1$ if the gumball is too small in.
- $s_3 = 1$ if the gumball is too large in diameter

The conveyor moves gumballs over a trap door that rejects defective ones. A gumball should be rejected if:

- It is too large ($s_3 = 1$), or
- It is both too light and too small ($s_1 = 1$ & $s_2 = 1$).

Implement the logic circuit that activates the trap door based on the sensor outputs.

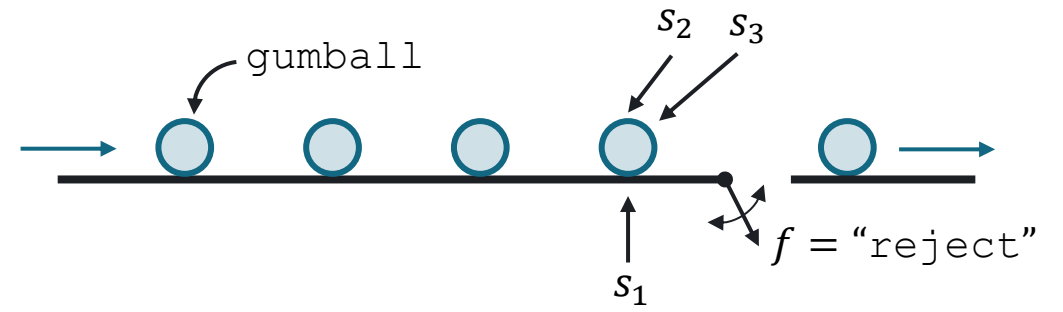
Conveyor and Sensors



EXERCISE

Solution

Conveyor and Sensors



EXERCISE

Implement the logic circuit that controls a single light in a large room with three entry points, each equipped with a switch. The behavior of this three-way light control are as follows:

1. The light is OFF when all three switches are open.
2. Closing any one of the switches turns the light ON.
3. If two switches are closed simultaneously, the light turns OFF.
4. If all three switches are closed, the light turns ON again.

Solution



LABORATORY

